

Shivani R Shukla, PhD

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I build in vitro platforms that read out glial and neuroinflammatory state electrically, at single-cell resolution, to study the mechanisms that drive electrode-tissue interface degradation.

EDUCATION

PhD in Bioengineering, University of California, San Diego | 2020-2025

BS in Materials Science & Engineering, Carnegie Mellon University | 2016-2020

Additional Major in Biomedical Engineering

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow | Biomimetronics | Chan Zuckerberg Biohub Chicago | September 2025-Present

- Novel fabrication methods for scalable nanoelectrode arrays using machine learning-guided two-photon polymerization, reactive ion etching, electron beam lithography, and interference lithography
- Topological routing for scalable, cross-platform bioelectronic design
- Flood filling networks for mixed hiPSC-derived brain cell cultures to assess neuroinflammation states on both immunofluorescent and electron microscopy datasets
- Develop a generalized Hodgkin Huxley model combined with pharmacological models for glial cell perturbations and functional assay development
- Creating multiphysics simulations for optimizing cell-electrode coupling in bioelectronic interfaces
- Design substrate-bound proximity assay using click chemistry for ion channel and GPCR sampling

PhD Candidate | Nanopillar Electrode Array Project | UC San Diego | 2021-2025

- Led design and fabrication of nanopillar electrode arrays achieving sub-threshold intracellular recording
- Demonstrated simultaneous electrophysiology and electrochemical sensing of neuronal networks
- Developed signal processing algorithms for extraction of multiscale electrical signals
- **Publications:** Nature Microsystems & Nanoengineering (2024), Advanced Materials Interfaces (2025), Small Science (2026)

PhD Candidate | Nanoscale Neural Interfaces | Forschungszentrum Juelich, Germany | 2024

- Implemented simultaneous patch clamp and nano/microelectrode array recording for validation studies
- Developed neuronal and synaptic guidance using nano-topographic chips
- **Publication:** Biosensors & Bioelectronics (2025)

PhD Rotation | TBI Detection Platform | UC San Diego | 2020-2021

- Designed multiplexed lateral flow assays detecting protease biomarkers after traumatic brain injury
- Developed behavioral testing protocol in mouse models to stratify TBI severity with 85% accuracy

Undergraduate Researcher | Graphene-Based MEA Project | Carnegie Mellon University | 2017-2020

- Developed nano-topographic graphene surfaces for enhanced cell adhesion and differentiation
- Designed self-rolling microelectrode arrays for 3D organoid electrophysiology
- Characterized neural/cardiac tissue interactions with nanocarbon interfaces
- **Publication:** Science Advances (2017)

SELECTED ACHIEVEMENTS

- Materials Research Science and Engineering Center (MRSEC) Graduate Research Fellowship (2024)
- Merkin Graduate Fellowship for Bioelectronics Research (2023)
- NSF Graduate Research Fellowship (2020)
- IEEE EMBS Student Chapter President, UC San Diego (2023-2025)
- Founder, Neuroengineering Journal Club at UC San Diego (2022-2025)

TECHNICAL EXPERTISE

Bioelectronics & Sensing: Nano- and microfabrication and characterization, Multi-modal bioelectronic interfaces (electrical/electrochemical), Signal processing for bioelectric measurements, Electrophysiology (patch clamp, calcium imaging, MEA recordings)

Biological Systems: Primary and iPSC-derived neuronal cultures, 2D and 3D neural network models, Cardiomyocyte and organoid systems, In vivo TBI models and assessment

Computational & Design Tools: Python, MATLAB for signal analysis, CAD design for bioelectronic devices, Machine learning for biological signal classification, Multiphysics modeling for bio-device interfaces, KiCAD